Executing the H-A0 to H-ANCHOR replacement: A-zero uniqueness as a sign-decomposition theorem modulo one anchor inequality

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

Scope (binding). This note executes H-A0 → H-ANCHOR + G-A0-DUI – a REPLACEMENT, NOT a full discharge. Full discharge requires the companion note that closes G-A0-DUI. The replacement holds AT THE PRODUCTION ANCHOR $\mu^2 = 0.005$ ONLY; the off-anchor neighbourhood is ROBUSTNESS-MU2 (OPEN). This closes the slaved isotropic $A = 0$ section, NOT global H-LAYER. Anchor constants remain verified dependencies under A1-KERNEL-CONV.

This note executes the operator-authorized replacement of the named hypothesis H-A0 by the weaker, explicit H-ANCHOR. The replacement rests on a sign-decomposition theorem (below) that derives H-A0's $A = 0$ uniqueness and zero-at-gap content from closed-form monotonicity lemmas plus the single anchor inequality $m^* > m_w$, now machine-verified. Scope: the slaved isotropic family of Math437 v1.2 (the $A = 0$ section of the layer). The replacement weakens the hypothesis burden of B2's and B1's T6; it performs no tier change on either (both stay T6 with a strictly weaker hypothesis).

2. What H-A0 asserted, and the reduction

H-A0 asserted (U) the $A = 0$ curve $F_0(m)$ has a unique minimum at the gap point m^*, (Z) of value 0 after subtraction, certified numerically on a consistent quadrature scheme (with a recorded 5.5×10^{-3} scheme-gap systematic). The sign decomposition below DERIVES (U) and (Z) from structure, so only the anchor inequality remains as an assumption.

With $r = \mu^2 = 0.005$, $u = -0.86$, $v = 3.24$, $M(m) = \int \frac{d^3k}{(2\pi)^3}[m + C(k^2 - q_0^2)^2]^{-1}$, $M_c = -u/(10v) = 43/1620$, $u_{\text{eff}}(M) = u + 10vM$, and the Math437 stationarity identity $dF_0/dm = \frac{1}{2}M'(m)\,g(m)$, $g(m) = r - m + 3uM + 15vM^2$ (so $g(m^*) = 0$ is the gap equation):

Lemma 1 (L1). *$M'(m) = -\int \frac{d^3k}{(2\pi)^3}[m + C(k^2 - q_0^2)^2]^{-2} < 0$ for $m > 0$ (differentiation under the integral; integrand and m-derivative dominated by integrable k^{-4}, k^{-6} tails for $m \geq m_0 > 0$ — the textbook residual G-A0-DUI).*

Lemma 2 (L2). *$g'(m) = -1 + 3u_{\text{eff}}(M(m))M'(m) \leq -1 < 0$ wherever $M(m) \geq M_c$ (i.e. $m \leq m_c$), since $u_{\text{eff}} \geq 0$ and $M' < 0$.*

Lemma 3 (L3). *On $M(m) < M_c$ (i.e. $m > m_c$), $u < 0$ and $0 \leq M < M_c$ give $g(m) < r + 15vM_c^2 - m = m_w - m$, $m_w := 0.0392407$; hence $g < 0$ for $m > m_w$.*

Theorem 1 (A-zero uniqueness, conditional on H-ANCHOR + G-A0-DUI)**.** *Assume L1 (modulo G-A0-DUI) and the anchor inequality $m^* > m_w$ (H-ANCHOR; with $M_R > M_c$ giving $m^* < m_c$). Then $g > 0$ on $(0, m^*)$ and $g < 0$ on (m^*, ∞) (L2 on $(0, m_c]$, L3 on (m_c, ∞), glued at $m_w < m^* < m_c$), so by L1 $F_0' < 0$ on $(0, m^*)$ and $F_0' > 0$ on (m^*, ∞): a strict global minimum at m^*, value 0 after subtraction. H-A0's (U) and (Z) hold as a theorem, with no curve-shape numerics and no quadrature-scheme dependence.*

3. The replacement

$$\boxed{\textbf{H-A0} \longrightarrow \textbf{H-ANCHOR} + \textbf{G-A0-DUI}}$$

where **H-ANCHOR**: at the production anchor $\mu^2 = 0.005$, the gap point satisfies $m^* > m_w$ (closed-form, $\times 7.76$ margin), with $M_R > M_c$ ($\times 4.12$); and **G-A0-DUI**: differentiation under the integral for $M(m)$ (textbook). H-ANCHOR is strictly weaker than H-A0 (H-A0 $\Rightarrow$ H-ANCHOR; the converse needs the derived lemmas): it is a single closed-form inequality, not a numerical curve-shape certification, so the quadrature-scheme dependence and the 5.5×10^{-3} scheme systematic EXIT the load-bearing chain (they remain only in the shared anchor PROVENANCE via A1-KERNEL-CONV). B2 and B1 hypothesis sets are updated $\{\ldots,$H-A0$\} \to \{\ldots,$H-ANCHOR$\}$.

Two clarifications (operator review). (a) $m_c \approx 19.96$ is NOT an operating mass: it is the monotone threshold solving $M(m_c) = M_c$, a long way above $m^* = 0.3045$, used only as the auxiliary boundary of the L2/L3 sign split. (b) G-A0-VER (the sign-decomposition arithmetic, m_w, $m^* > m_w$) is CLOSED here; G-A0-DUI (the analytic regularity $M' < 0$) is a residual closed only in the companion ga0-dui-closure note — so "G-A0-VER closed" and "G-A0-DUI residual" coexist without contradiction, and this note is the EXECUTION of the replacement, the companion note the COMPLETION of the residual.

4. Numerical verification

Script `codes/vacuum/ha0_sign_decomposition.py` (v1.0.0; 14/14 self-test asserts PASS; artefact `claims/B2-PROPA-HLAYER/runs/260606-ha0-sign-decomposition/result.json`). G-A0-VER is CLOSED.

Quantity	Value	Check
$M_c = -u/(10v)$	$43/1620 = 0.0265432$	closed form
$m_w = r + 15vM_c^2$	0.0392407	closed form
gap equation $g(m^*)$	2.6×10^{-14}	≈ 0; $m^* = 0.304526$
H-ANCHOR $m^* > m_w$	$0.30453 > 0.03924$	$\times 7.76$ margin
$M_R > M_c$	$0.109414 > 0.026543$	$\times 4.12$ margin
L1 $M' < 0$ / L2 $g' \le -1$ / L3 $g < m_w - m$	all hold; $m_c = 19.96$	sign lemmas verified
F_0' sign structure	< 0 below m^*, > 0 above, 0 at m^*	unique minimum

Quantitative sanity check (mandatory): g and m share the dimension of $r = \mu^2$; the gap identity $m^* = r + 3uM_R + 15vM_R^2 = 0.304523$ reproduces the certified $r_R = 0.3045257$ to 5×10^{-6} (display rounding of M_R), confirming $m^* = r_R$ by the gap equation, so the anchor inequality is on the same footing as the rest of the certified chain.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3.5; binding gate, no external review this turn)

(α) *"G-A0-DUI is still a gap — is it really textbook?"* VALID-with-mitigation: $M'(m) = -\int[m + C(k^2 - q_0^2)^2]^{-2}$ is the Leibniz rule under dominated convergence; for $m \geq m_0 > 0$ the integrand ($\leq m_0^{-1}$ near shell, k^{-4} tail) and its m-derivative (k^{-6} tail) are dominated by fixed integrable functions. Standard; registered as G-A0-DUI, expected routine. The dominating functions are stated, so a reviewer can close it directly.

(β) *"Is m^* a hypothesis or a fact?"* DISMISSED: m^* is the gap solution at $\mu^2 = 0.005$, a DEPENDENCY value (gap equation + A1-KERNEL-CONV convention), not a free assumption. H-ANCHOR asserts the INEQUALITY $m^* > m_w$, a verified closed-form fact ($\times 7.76$). Anchor-value provenance is a dependency, not a hypothesis.

(γ) *"Is H-ANCHOR genuinely weaker than H-A0 or just relabeled?"* DISMISSED: H-A0 asserted the entire curve shape (uniqueness + zero-at-gap + monotone flanks) numerically; H-ANCHOR asserts ONE inequality, with the rest DERIVED by L1/L2/L3. Strictly weaker (H-A0 $\Rightarrow$ H-ANCHOR, not conversely); the 5.5×10^{-3} scheme systematic exits the chain.

(δ) *"Does the swap break B1's or B2's T6?"* DISMISSED: it IMPROVES them — both now rest on a strictly weaker hypothesis. Hypothesis lists update H-A0 $\rightarrow$ H-ANCHOR; H-ANCHOR is registered in GATES.md; tier-monotonicity unaffected (linter PASS).

(ϵ) *"H-ANCHOR is anchor-pinned to $\mu^2 = 0.005$ — robustness?"* VALID-with-mitigation: like H-A0 before it, H-ANCHOR is stated at the production anchor; the μ^2-neighbourhood is the pre-existing ROBUSTNESS-MU2 gate, unchanged by the swap. The swap does not worsen it.

No objection UPHELD as a blocker (3 dismissed, 2 valid-with-mitigation, each a textbook residual or a pre-existing tracked gate).

6. Result footer

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Result ID:              B2-PROPA-HLAYER / H-A0 -> H-ANCHOR replacement
Precise statement:      A-zero uniqueness + zero-at-gap (H-A0's U,Z)
                        derived as a sign-decomposition theorem:
                        F_0'(m) = (1/2) M'(m) g(m) with M'<0 (L1), g
                        strictly decreasing on the window (L2), g<0
                        beyond it (L3), unique zero g(m*)=0 -> unique
                        minimum at m*, conditional only on H-ANCHOR
                        (m* > m_w, x7.76) and the textbook G-A0-DUI.
                        H-A0 replaced by H-ANCHOR (+ G-A0-DUI).
Scope:                  Slaved isotropic A=0 family (Math437 v1.2);
                        production anchor mu^2 = 0.005. Quadrature-
                        scheme curve-shape certification removed.
Dependencies:           A1-KERNEL-CONV (anchor provenance); the gap
                        equation. B2/B1 carry H-ANCHOR forward.
Evidence grade:         ANALYTIC (sign lemmas + theorem) + EXECUTED
                        (ha0_sign_decomposition.py 14/14)
Reproduction command:   python codes/vacuum/ha0_sign_decomposition.py
```

Expected output:	claims 14/14 PASS; m*/m_w = 7.76, M_R/M_c = 4.12; result.json under claims/B2-PROPA-HLAYER/runs/ 260606-ha0-sign-decomposition/
Falsification gate:	An anchor with m* <= m_w (refutes H-ANCHOR); M’ >= 0 somewhere (refutes L1); a second zero of g on (0,inf) (refutes the sign decomposition).
Tier before / after:	B2 T6 / T6 (hypothesis H-A0 -> H-ANCHOR, strictly weaker); B1 T6 hypothesis set likewise updated. No tier number change; the theorems strengthen.
No-overclaim statement:	The replacement weakens the hypothesis; it does NOT discharge it (H-ANCHOR + G-A0-DUI remain). G-A0-DUI is textbook but not yet written out; ROBUSTNESS-MU2 (off-anchor) remains open and is unchanged. No tier-number change on B1/B2.
Next required action:	Write out G-A0-DUI explicitly (close the textbook residual) to discharge H-ANCHOR toward absorption into the A1 convention dependency; ROBUSTNESS-MU2 for the mu^2-neighbourhood.